

Nine exchanges, one rule: spend only what is already here

Every settlement runs one exchange. A client who wants to trade on another world first moves money there through the official backbone, and from then on that exchange trades for the client using only balances it already holds. Nothing is ever spent on the strength of a message that might still be in flight, so lost, late or duplicated packets can delay the market but cannot create or destroy a single NeoDollar or share.

This paper states the architecture and the operating rules (Sections 1–2), the guarantees we give and what they cost (Section 3), the weaknesses we found by attacking the design (Section 4), how it keeps working indefinitely (Section 5) and what a real deployment would change (Section 6). The evidence appendix holds the traces and calculations behind every number; a full discrete-event simulator of the brief's network and our ledgers reproduces all of them. Headline results at the 2026-09-22 epoch:

- **Earth buyer, Mars shares:** funding, trade and delivery of 1,000 Ares shares home in 3.07 h with no loss (47 packets in total); median 3.07 h and 99th percentile 7.70 h over 2,000 random-loss runs, every run completed.
- **300-hour capped futures:** \$160,000 of margin (32% of all cash) locked at Mars for the whole contract; both price paths settle with both payouts spendable at home by 328.17 h. The rejected oversize variant ($Q = 2,501$) leaves everyone whole.
- **All nine settlements** can fund, trade and receive payouts. Neptune is the hardest: its direct packets are lost 91% of the time, so a trade there completes within 72 h with probability 0.47 and within 168 h with probability 0.97.

1. Architecture

Institutions. There are exactly nine institutions: one exchange per settlement (the brief allows twelve). There is no clearinghouse. Each exchange keeps customer accounts in its own ledger and is the only writer of those balances. The three stock markets sit with their share registries and their opening sellers: Ares Habitat on Mars, Belt Works on Ceres, Terra Fabrication on Earth. The capped futures contract is hosted by the Mars exchange because Mars is the best-connected settlement for the inner system (Appendix S3) and the price source, Bob, lives there.

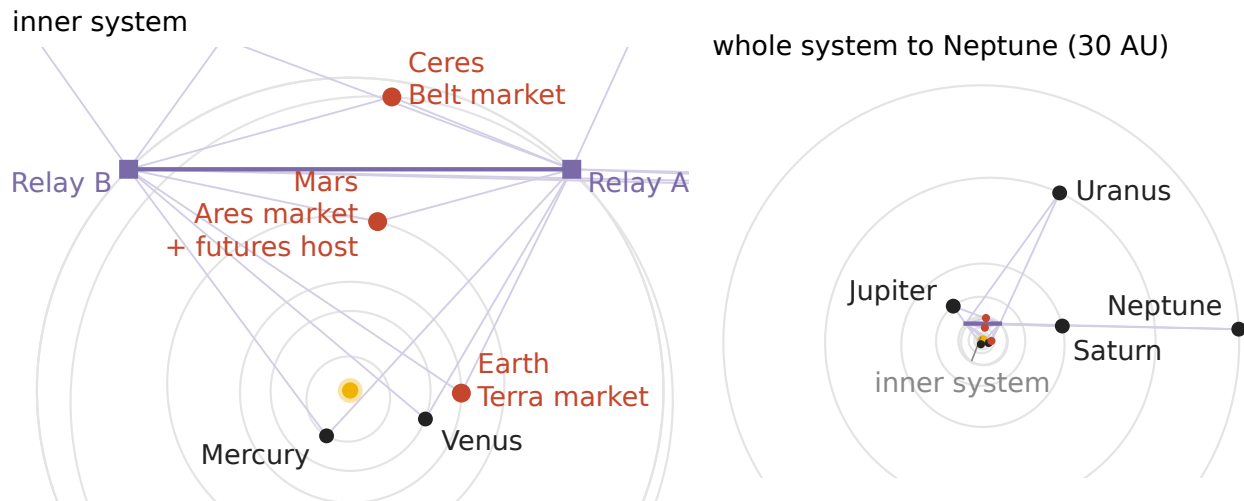


Figure 1. The nine settlements and two relays at the epoch. Purple lines are the 19 backbone links. Red settlements host a market. The geometry is never a straight line in practice: every flight is solved with the moving receiver, and links shut whenever a path passes within 0.1 AU of the Sun.

Accounts. A client has one registered home and may hold a trading account at any of the nine exchanges. Remote accounts start empty and only ever receive assets by an explicit transfer. Opening a remote account creates no wealth, no new identity and no extra direct-service quota.

Shared state	The only institution allowed to change it
Available balances, order holds, contract escrow	The exchange holding those balances
Order book and every fill	The asset's market exchange
Debit and terms of an outgoing transfer	The sending exchange
Credit of an incoming transfer	The named receiving exchange (exactly once)
Share registry: which exchange holds how many shares	The asset's registry exchange (= its market)
Contract position, fixing and payout allocation	The contract host (Mars)

Opening balance sheet (hour 0, identical for every run): Alice (Earth) \$150,000; Bob (Mars) \$50,000 and 3,000 Ares; Cara (Ceres) \$100,000 and 1,000 Belt; Dax (Neptune) \$100,000; Eve (Uranus) \$50,000; Fin (Earth) \$50,000 and 1,000 Terra. Total \$500,000 and 5,000 shares in six accounts at five settlements. Institutions start with zero.

How a cross-planet trade works

1. **Fund.** Alice tells Earth (local access, 1 s) to move \$100,000 to her Mars account. Earth debits her at once and sends a TRANSFER record over the backbone. While in flight, the money belongs to Alice but nobody can spend it.
2. **Import.** Mars accepts that transfer exactly once, credits Alice's Mars account, and sends RECEIVED back.
3. **Order.** When Earth has the receipt, Alice knows her money is at Mars and sends a signed order by the direct service (several copies, because direct packets are often lost).
4. **Trade.** Mars holds \$100,000 of Alice's cash, matches Bob's resting sell order and swaps cash for shares atomically. Bob's money is spendable on Mars at that instant.
5. **Deliver.** Alice chose home delivery, so the same atomic step exports her 1,000 shares to Earth. The trade is complete for Alice only when Earth credits them.

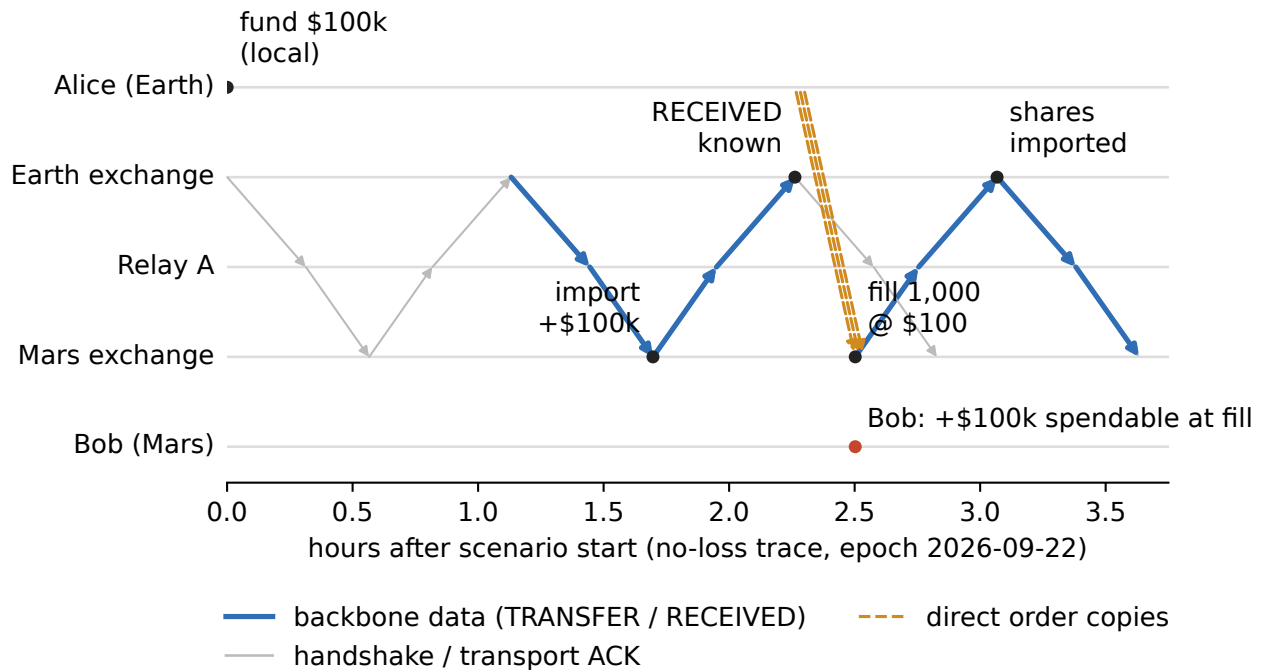


Figure 2. The no-loss Earth-Mars trade (scenario S1a). Bold arrows carry money or shares; grey arrows are the brief's handshake and acknowledgement traffic; dashed arrows are Alice's three direct order copies. Complete at 3.067 h.

2. Operating rulebook

Each rule names who acts, what they may rely on, and what happens when a message goes wrong. "Durable" means the change survives a crash or reset; every change below is one atomic durable write.

R1. Moving assets between exchanges

A TRANSFER carries a globally unique ID (source exchange + export number), the owner, asset, amount, destination and purpose; its terms never change. (1) The source checks the owner's *available* balance and debits it in the same write that creates the export, so the value now exists only as "in transit". (2) The destination credits it the first time it sees that ID and records the ID forever; any later copy only triggers another RECEIVED. (3) The source marks the export delivered when RECEIVED arrives. A missing receipt **never** refunds the source: the courier keeps re-sending the same record (hop retries, endpoint retries, then application resubmission after 24, 48, 96 and 168 h) until the receipt comes back. A transfer to an account that has been closed is credited to a restricted balance and sent straight back with a new transfer. Shares always move between the registry exchange and a custodian, so the registry can reconcile its custody totals.

R2. Orders, holds and matching

An order names its owner, a unique ID, asset, side, integer quantity and limit price, time-in-force (GTC or IOC), an expiry no more than 30 days away and a disposition (keep results at the market, or send them home automatically). The market admits it only if the owner's *available* balance at that exchange covers it: quantity $\times$ limit for a buy, the shares for a sell. Those assets move into a hold that nothing else can touch. An order that is not covered is rejected for good; the client must fund the account and send a new order ID. Matching is price-time priority at the resting price; self-trades are skipped. Each fill is one atomic write: buyer hold $-q$ ·limit, seller hold $-q$ shares, buyer $+q$ shares and $+q$ ·(limit – price) refund of price improvement, seller $+q$ ·price, plus the automatic home export when chosen. Cancels release only the unfilled remainder; a cancel that arrives

before its order leaves a tombstone so the late order can never open. There is no amend: cancel and send a new order.

R3. Withdrawals and account closure

WITHDRAW names an amount or ALL. ALL is evaluated once, when the host first processes it, and repeated copies return that same export. A withdrawal can only take available balance; it never cancels orders or touches margin. CLOSE stops new orders, cancels open ones and withdraws what is free; escrow stays until the contract pays out, and those payouts are then sent home automatically. The home exchange can never "unlock" money that sits at another exchange: only the host that holds it can send it back.

R4. Capped futures and the one margin rule

Contract. Long and short agree on Q NeoDollars per index point, strike $K = 100$, and a collar $[60, 140]$. The long's lifetime payoff is $Q \times (\text{clip}(P, 60, 140) - 100)$; the short receives the negative.

Margin rule (stated before any price is seen). Each side must hold its *maximum possible loss*, $Q \times 40$, in escrow at the host from opening until fixing. Nothing else: no variation margin, no calls, no offsets, no share collateral. Because the escrow equals the largest loss either side can ever owe, no price path can produce an unfunded obligation, and the rule is identical in the rising and falling runs.

Each party funds its Mars account (R1), then sends a contract instruction carrying the full terms. Mars moves that party's margin from available cash into a hold; when both holds exist the contract opens atomically and maturity is set to opening + 300 h. If both are not held by the opening deadline (168 h after the scenario starts) the contract is rejected and each hold returns to that party's available Mars balance; clients who hear no OPEN report then withdraw. Payouts are exported home automatically in the same write as the fixing.

R5. Price source and fixing

Bob, the named index source at Mars, learns the schedule locally at opening and publishes one signed observation at opening + 60, 120, 180, 240 and 300 h by local access. Mars fixes at maturity + 24 h using observation 5; if it is missing or two conflicting copies exist, it uses the latest earlier valid observation, else 100. Late observations are ignored. The 24 h grace lets a remote source's packet arrive before fixing; with a local source it costs one day of escrow (3.84 M dollar-hours) and could be cut (Section 4).

R6. Which service carries what

Message	Service	Quota charged
TRANSFER (funding, withdrawal, auto-delivery, payout), RECEIVED	Backbone, pinned route of ≤ 3 links	Sending exchange, 66 per rolling 24 h
Order, cancel, withdraw, close, contract instruction	Local access at the host, else direct	Client's own 12 per 24 h, ≥ 60 s apart
Price observation	Local access at Mars (Bob is local)	none
Contract status REPORT to a remote client	Direct	Mars exchange principal, 12 per 24 h

The nine exchanges split the 600 backbone originations as 66 each (594 total); there is no shared counter. SYNs, first transmissions, RECEIVEDs and application resubmissions count; automatic retries and ACKs do not, but every launch is counted in our packet totals. Client instructions never ride on the backbone, and exchange coordination never rides on the direct service.

R7. Declared client policy

Clients are software with the published ephemeris but no knowledge of hidden incidents.

- **Fund first, then instruct.** A client sends an order or contract instruction only after its home exchange has recorded RECEIVED for the funding transfer. It learns this by local access, so the order cannot be rejected for missing funds.
- **Copies.** Send the smallest number of direct copies, 60 s apart, that gives $\geq 99\%$ chance at least one arrives, capped at 8 (Earth→Mars: 3 copies, all lost with probability 0.0021; Neptune→Mars: 8 copies, 0.46).
- **Fallback.** An equity buyer whose shares have not arrived home by twice the direct + backbone delay plus 6 h re-sends the same order ID (duplicates are harmless) until the order's own expiry. A futures party that has no HELD/OPEN report after twice the direct flight plus 6 h re-sends its instruction, until the deadline.
- **Access check.** Before funding, the client checks that its direct path to the host stays open for the whole planned sequence. If not, it waits for the next long-enough window, or, if that is after the deadline, suspends; its money stays at home.

R8. Recovery

Ledgers, outboxes and ID records are durable. After a reset, an expired session or a delivery the transport gives up on, the courier opens a new session and resubmits the same records after 24, 48, 96 and then every 168 h, waiting out known closures and quota. Recovery work is preferred over new work.

What each actor knows when a message goes wrong

Event	Sender knows / does	Receiver knows / does	Who owes what
TRANSFER lost or late	No RECEIVED yet; keeps retrying the same ID	Nothing; the money cannot be spent there	Value is "in transit", owned by the client, backed by the source debit
RECEIVED lost	Retries the TRANSFER	Duplicate import ignored, RECEIVED re-sent	Nothing changes; money already spendable at host
Duplicate TRANSFER / order / instruction	—	Same ID and terms: same result returned; changed terms: rejected	No double effect
All order copies lost	No shares home by fallback time: re-sends same ID	No order exists; cash stays available	Client's cash idle at host
Cancel lost or late	Assumes nothing until state confirmed	Order may still fill	Fill stands; cancel releases only the remainder
Contradicting price copies	—	Observation invalid; fall back to earlier one	Fixed at cutoff; never revised
Host isolated 72 h	Remote clients cannot instruct	Local trading, matching, maturity continue	Payout exports wait, funded, in the outbox
Exchange reset	Volatile sessions lost	Ledger intact; outbox resubmitted after backoff	No balance changes

Option O2 Forwarding through a third exchange

Not part of the v3 baseline; offered for adoption. When the client's direct path to the host is blocked by the Sun but both legs through some third settlement are open, the client sends its signed instruction to that settlement's exchange, which re-sends it by direct service under *its own* 12-per-day quota. The message stays client communication; the forwarder gains no authority and the host still checks the client's signature and balances. At the 2028 Mars conjunction this turns "no Earth–Mars trading for 53 days" into a 4.55 h trade (Appendix E5).

3. Guarantees, conditions and costs

We separate what the **rules** guarantee in every possible run from what the **model** (orbits, loss law, quotas) makes likely. The first kind holds for any packet loss, delay or order; the second is evidence, not proof.

#	Guarantee	Holds because	Condition
G1	Total NeoDollars and each share type never change	Every write debits and credits equal amounts; transit is a ledger state	Rules only. Checked after every event in every traced run
G2	No asset serves two uses at once	Available, held, escrowed and in-transit are disjoint states	Rules only
G3	Every futures payout is fully funded	Escrow = maximum loss, posted before opening	Rules only; any price path
G4	Each transfer is credited exactly once	Destination remembers every export ID forever	Rules + durable storage
G5	A host fill is final and spendable at the host at once	Both holds exist before the atomic fill	Rules only
G6	A funded transfer eventually arrives	Retries never stop and the backbone is always connected (E4)	Model: no unconditional deadline exists under random loss
G7	Completion times	Measured	Model: see probabilities below and in Appendix E2/S3

Service table. Every product is offered at every settlement: funding/withdrawal at all three markets, equity trading in Ares, Belt and Terra, and the capped futures at Mars. What differs is speed and probability. Columns: one-way backbone delay to the Mars/Ceres/Earth hosts (minutes, hour 0); direct copies per instruction to Mars and the chance at least one arrives; and for a \$100k Ares purchase delivered home: no-loss time / median under loss / probability done within 72 h (1,000 runs each, Appendix S3).

Settlement	Backbone min	Direct copies → P	Ares trade h: no-loss / median / P(≤72 h)	Restrictions
Mercury	42/42/46	3 → 0.997	3.8 / 3.8 / 1.00	Direct path to Mars closed h 22–357 at epoch: funding waits (R7) or O2
Venus	37/36/41	3 → 0.997	3.3 / 3.3 / 1.00	Closed near Venus–Mars conjunctions; waits or O2
Earth	34/33/loc	3 → 0.998	3.1 / 3.1 / 1.00	Earth–Mars direct closed ~53 days per 2.1 y (Mars conjunction); O2
Mars	loc/30/34	local	0.0 / 0.0 / 1.00	All products local, 1 s
Ceres	30/loc/33	2 → 0.992	2.6 / 2.6 / 1.00	Belt market local
Jupiter	40/41/51	4 → 0.992	3.9 / 4.0 / 1.00	4 copies
Saturn	76/75/79	7 → 0.990	7.6 / 10.6 / 1.00	7 copies
Uranus	155/154/158	8 → 0.885	15.4 / 26.8 / 0.99	8 copies; fallback often used
Neptune	247/246/250	8 → 0.545	24.7 / 74.4 / 0.47	8 copies still lose 46% of bursts; slow

All futures parties must reach Mars before the 168 h opening deadline. Inner-system parties almost always do (Appendix E2: the contract opened in every one of 2,000 lossy runs). For the futures, a Neptune party (Dax long against Cara) gets the contract opened in 99% of lossy runs before the 168 h deadline; the rest are rejected at the deadline with all margin returned (Appendix S3).

Costs. A one-off cross-planet trade with home delivery costs 47 packets (44 backbone launches including handshakes, receipts and ACKs, plus 3 direct copies) and 5 backbone originations. Repeat trading is far cheaper: the \$10k worked example (S1i) made four fills, one cancellation, a cancel/fill race and a final sweep

home on one funding transfer, 59 packets in total. Capital: the futures lock \$160,000 for 324 h (escrow 51.84 M dollar-hours, peak encumbered / notional = 0.80); an equity purchase holds 100% of its value while in transit or on order and nothing afterward. No institution holds capital, and none is needed.

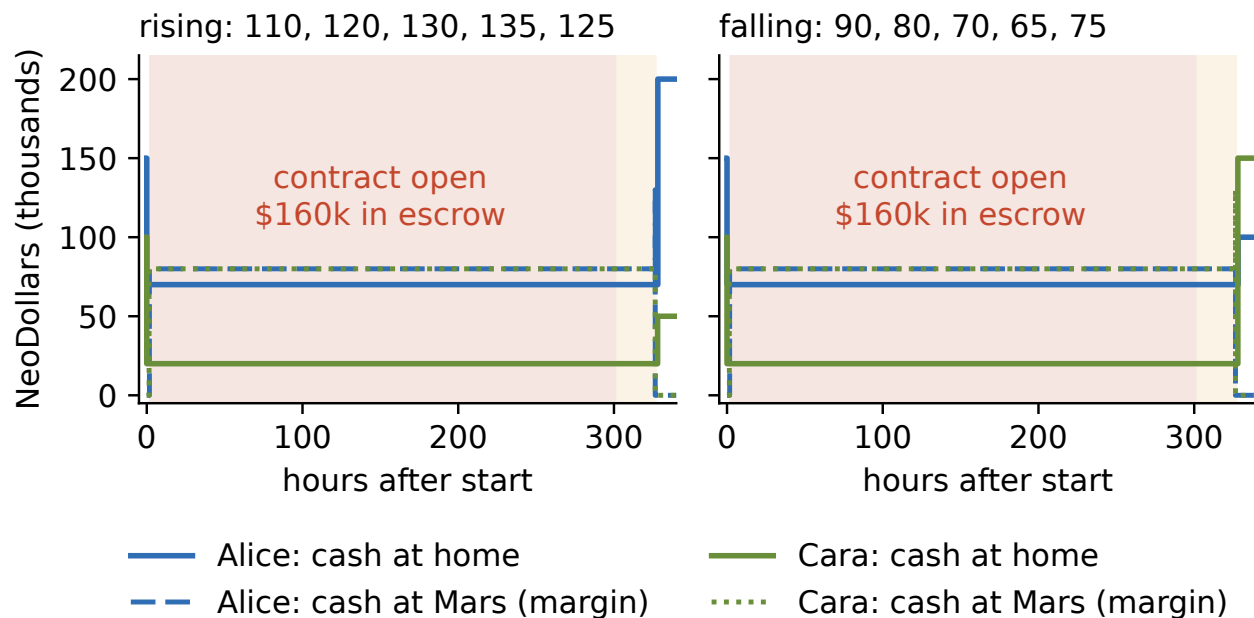


Figure 3. Where Alice's and Cara's cash is during the futures (no loss). Shaded: contract open (red), fixing grace (beige). Both paths use the same margin and finish with \$500,000 in total.

4. Risks we found, ranked

Ranked by harm $\times$ likelihood under our own evidence. The interesting ones are interactions between our own rules and the geometry; they were found by the stress search, the shifted epochs and the outer-settlement runs, not by inspection.

#	Risk	Evidence	Treatment
1	Solar conjunction cuts a client off from its host	E5: at the 2028 Mars conjunction the Earth–Mars direct path is shut for 53 days. No Earth client can order, cancel or withdraw at Mars; Bob's 30-day sell order expires first, so the trade never happens	Access check before funding (money stays home); futures suspend safely. O2 forwarding restores service: trade in 4.55 h, both futures runs complete
2	Rule interaction: funds parked at a host become unreachable	Withdrawals are client instructions (direct), so money left at Mars during a closure cannot be called home even though the backbone works	Clients fund just in time (R7); automatic home delivery and payout need no instruction; O2
3	Outer settlements: direct loss 76–91% per copy	S3: Neptune $P(\text{done} \leq 72 \text{ h}) = 0.47$; 8 copies give only a 54% burst	Improvement I1 (applied): copies capped at 8 not 6, fallback after 6 h not 24 h, resend until order expiry: Neptune 0.34 $\rightarrow$ 0.47, Uranus 0.93 $\rightarrow$ 0.99
4	Host concentration	All futures margin and the price source are at Mars. An isolation of Mars around fixing delays both payouts by up to 75 h (S2 search).	Fixing and payout are local and automatic; payouts wait funded in the outbox; nothing needs a remote instruction

#	Risk	Evidence	Treatment
5	Rule interaction: access check vs. closures of the "wrong" path	S3 found Mercury's funding stuck for 335 h with a 48 h check window, because Mercury → Mars direct is shut h 22–357	Improvement I2 (applied): the check covers the real sequence length; Mercury now completes in 3.81 h
6	Silent session loss after an endpoint reset	S2 (worst of 56,280 incidents): a Mars reset at h 0.75, just after the handshakes, makes Mars drop the funding transfers silently; senders wait ~100 h of endpoint retries + 24 h backoff. Futures done 453.79 h instead of 328.17 h (still opens inside the deadline; no money at risk)	A1 restart notice to recent peers: done 329.49 h. Recommended
7	Shared host REPORT quota	Mars sends every remote client's status by direct service on one 12-per-day quota	Reports only on state change, ≤ 1 per 6 h per client; clients rely on fallbacks; demand cap in Section 5
8	Single price source	Bob can fail or publish conflicting copies	Deterministic fallback to latest valid earlier observation, else 100; disclosed basis risk
9	Custodian trust	Prefunding concentrates exposure on the host exchange	Brief assumes operators follow rules; independent audit is future work

5. Long-term operation

The backbone never disconnects (geometry, proven). The relays are 90° apart on one circular orbit and the A–B link always clears the Sun by 2 AU. A gateway–relay link is blocked only when the gateway sits nearly behind the Sun as seen from that relay (an angle of at least 159°), which cannot happen for both relays at once. So every settlement always has a route of at most three links to every other, outside maintenance and incidents. The 200-year scan confirms it: every settlement had a route to every market host at all 1,753,201 hourly samples, and the slowest one-way route ever seen was Mars→Neptune at 315 min. Direct paths, by contrast, do close for weeks (conjunctions), so every direct-dependent promise is conditional.

Long periods with no usable route, and the backlog after. Nothing has a wall-clock expiry except orders (≤ 30 days, releasing only their own hold) and contract opening deadlines (returning margin). Transfers, receipts and payouts wait in durable outboxes; sessions that idle for 7 days are simply re-opened; a packet that hits the 30-day transport lifetime is replaced by an application resubmission of the same financial ID. After a route reopens, each exchange drains its outbox within its 66-per-day origination budget, recovery first (22 of the 66 are reserved for it). Up to 5 transfers share one 1,024-byte packet, so a backlog of 300 transfers to one peer drains in about a day.

Obligations past our longest example. A contract stores absolute maturity and fixing times; fixing is a local computation at the host and the payout export is created in the same write, so a maturity years away needs no message to happen.

Capital. There is no endowment. New capital can only enter through fees or contributions debited from real accounts and moved by R1. Escrow is released only by fixing or rejection, so capital cannot leak or be invented.

Storage and identifiers. IDs are 96-bit counters behind a 32-bit principal prefix: at a million IDs per second they last 2.5×10^{15} years. Time is kept as signed 64-bit integer milliseconds (≈ 292 million years each way). Our simulator uses float seconds, which stop resolving 1 ms after 142,710 years; a production system must use integer time and extended-precision orbital phase. Bounded state: ≤ 32 live orders per client per host, 256 per host, 4,096 unresolved transfers per host with 20% reserved for returns and maturities. Terminal records are compacted behind per-principal floors; an unresolved one blocks compaction and, eventually, new work. When a limit is reached we stop accepting new work; we never drop an obligation.

Demand we accept. Per exchange per day: up to 44 routine backbone originations (≈ 10 new cross-exchange funding or withdrawal workflows at 4–5 originations each) and 22 recovery originations; per client, 12 direct packets, i.e. 1–4 remote instructions depending on distance. Above that, work queues; nothing is promised for it.

Eons. The brief's orbits are fixed ellipses, so the geometry is quasi-periodic forever: the 200-year statistics describe every later century of the model. What ages is arithmetic, not orbits. With integer milliseconds and extended-precision phase, the rules above have no time-dependent part; the binding limits are identifier space (above) and the physical infrastructure, which the brief fixes. The real Solar System would drift away from these elements within centuries (Section 6), so we do not claim the model's geometry beyond its own terms.

6. Deployment assessment

The baseline leaves real effects out on purpose. None of the following can break conservation or double use (they only delay or lose packets, which the rules already tolerate); they would change our times and probabilities.

Effect	Size estimate	Source and assumptions	Effect on guarantees
Earth rotation hides the sky	≈ 11.97 h below-horizon gaps per station	Sidereal day 23.93 h; one equatorial station, zero elevation mask [1]	Times only: +up to 12 h per hop without a 2nd station. G1–G5 unchanged
Doppler shift	≈ 3.20 MHz at 32 GHz	f-v/c with an assumed 30 km/s line-of-sight speed [1, 2]	Modems must track it; if not, more loss. Loss only slows us
Clock rates (relativity)	Mercury vs Earth clocks: $\approx 2.3 \times 10^{-8}$, ≈ 0.7 s/yr	$GM/(rc^2) + v^2/(2c^2)$ at 0.39 and 1 AU	Maturity and fixing use TDB; local clocks must be corrected, else ms-level schedule errors
Real ephemeris drift	Not estimated	Kepler elements ignore planetary perturbations	All timetables need a live ephemeris; rules do not

[1] NASA Earth Fact Sheet, nssdc.gsfc.nasa.gov/planetary/factsheet/earthfact.html. [2] ESA, Cebreros DSA-2 ground station (Ka-band, 32 GHz). [3] CPMI–IOSCO, Principles for Financial Market Infrastructures (delivery versus payment). [4] Gray and Lamport, Consensus on Transaction Commit, 2004.